$$x + 75° = 180°$$
$$x = 180° - 75°$$
$$x = 105°$$

Example 4.3 In ΔSTU, if $SU = UT$, $\angle SUT = 70°$, $\angle STU = x$, find the value of x.

Solution

Given, $\angle SUT = 70°$

$\angle UST = \angle STU = x$ [Angles opposite to equal sides]

$\angle SUT + \angle UST + \angle STU = 180°$

$70° + x + x = 180°$

$70° + 2x = 180°$

$2x = 180° - 70°$

$2x = 110°$

$$x = \frac{110°}{2} = 55°$$

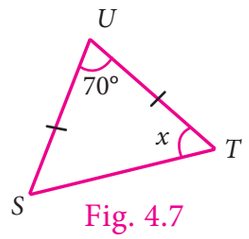


Fig. 4.7

Example 4.4 If two angles of a triangle having measures 65° and 35°, find the measure of the third angle.

Solution

Given angles are 65° and 35°.

Let the third angle be x

$65° + 35° + x = 180°$

$100° + x = 180°$

$x = 180° - 100°$

$x = 80°$

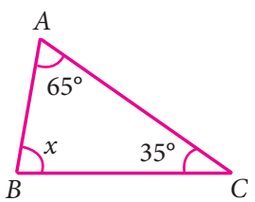


Fig. 4.8

4.3 Exterior Angles

We know that any triangle is made up of three vertices, three sides and three angles. Now observe the triangle given in Fig. 4.9

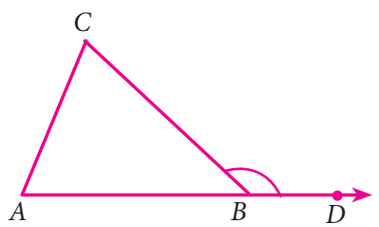


Fig. 4.9

In ΔABC, the side AB is extended to D. Observe $\angle CBD$ which is formed by the side BC and BD. $\angle CBD$ is known as the exterior angle of ΔABC at B.

We can observe that, $\angle ABC$ and $\angle CBD$ are adjacent and they form a linear pair.

Think

Can CB be extended to F to get another exterior angle of ΔABC at B?

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